

Major Applications

- General Purpose Relays

- Rated by the amount of current its switch contacts can handle
- Most versions have one to eight poles and can be single or double throw
- Computers, copy machines, appliances, Security, Pool & Spa, Industrial Controls, Office Equipment



- Power Relays

- Capable of handling larger power loads – 10-50 amperes or more
- single-pole or double-pole units
- Office Automation, Process Control, Automotive, Energy Management Systems, Motor Drives/Controls, Appliance Controls

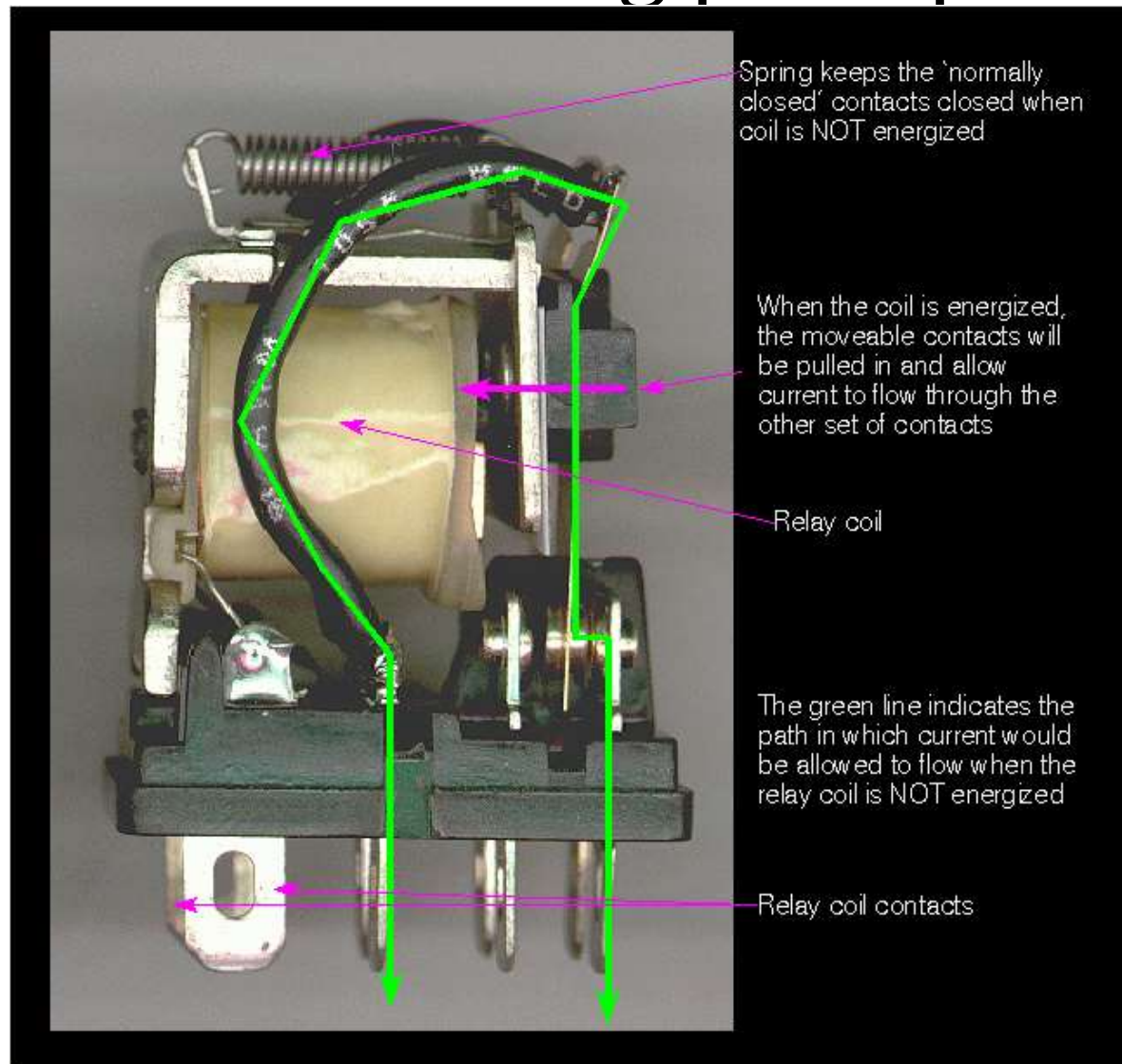


Major Applications

- Contactors
 - Control high voltages and currents in industrial electrical applications
 - Automotive, Control Systems, overload circuitry, fluid systems
- Time Delay Relays
 - Delay on operate
 - contacts might not open or close until some time interval after the coil has been energized
 - Delay on Release
 - contacts will remain in their actuated position until some interval after the power has been removed from the coil
 - Interval timing
 - contacts revert to their alternate position at a specific interval of time after the coil has been energized.
 - Flashing light control, Engine auto start control, Conveyor belt sequence delay, Motor soft-start delay control, Furnace Safety purge control

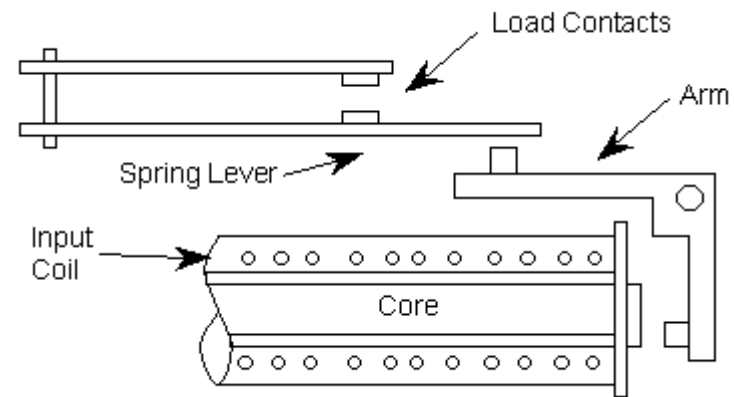
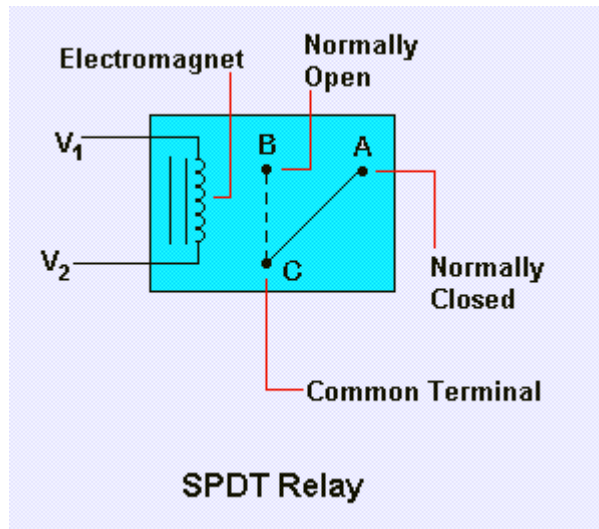


Basic working principle



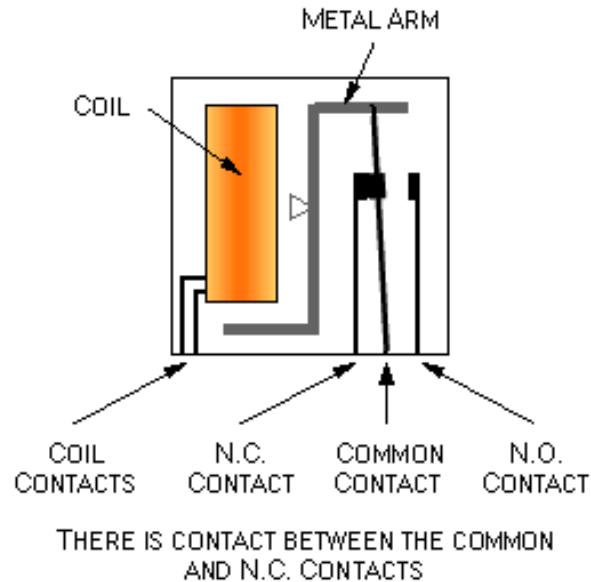
Basic working principle

The figures below demonstrate the typical electro-mechanical relay (EMR). An input voltage is applied to the coil mechanism. The input voltage magnetizes the core which pulls the arm towards it. This action causes the output contacts to touch, closing the load circuit. When the input voltage is removed, the spring lever will push the contacts away from each other, breaking the load circuit connection

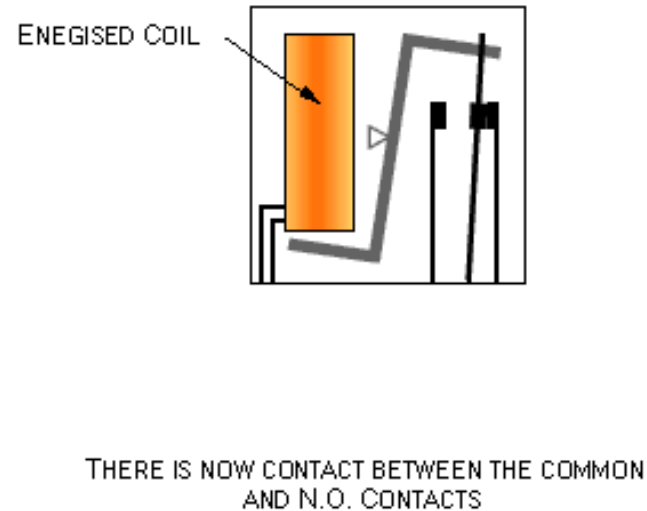


Basic working principle

2a: Relay off

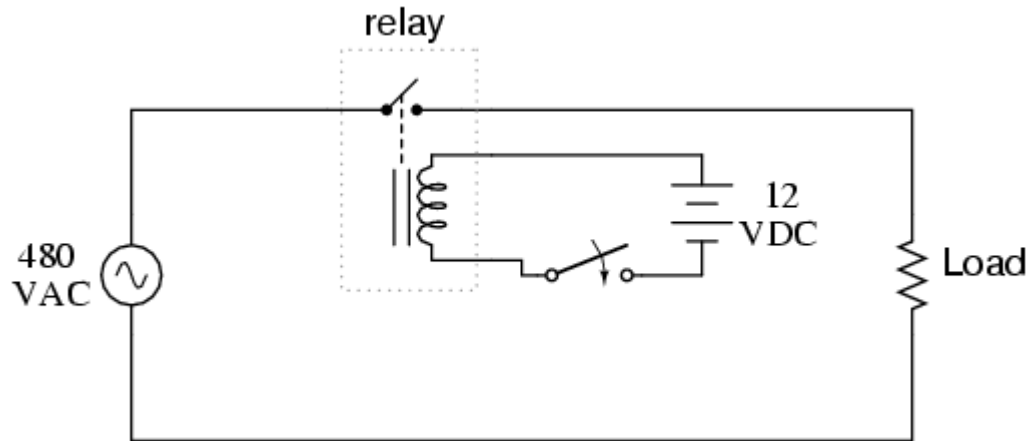


2b: Relay on



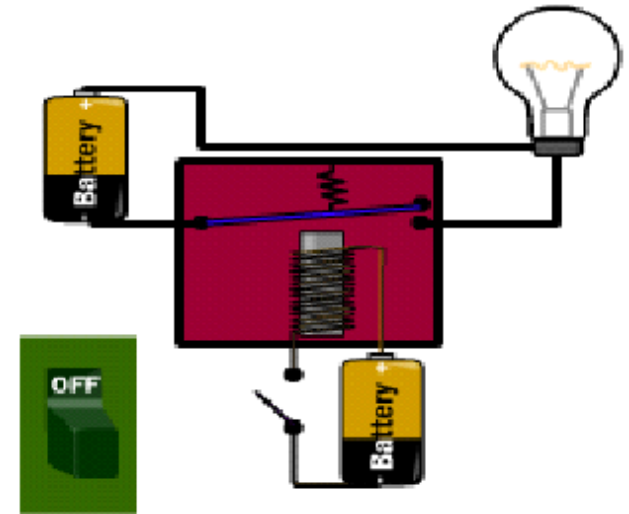
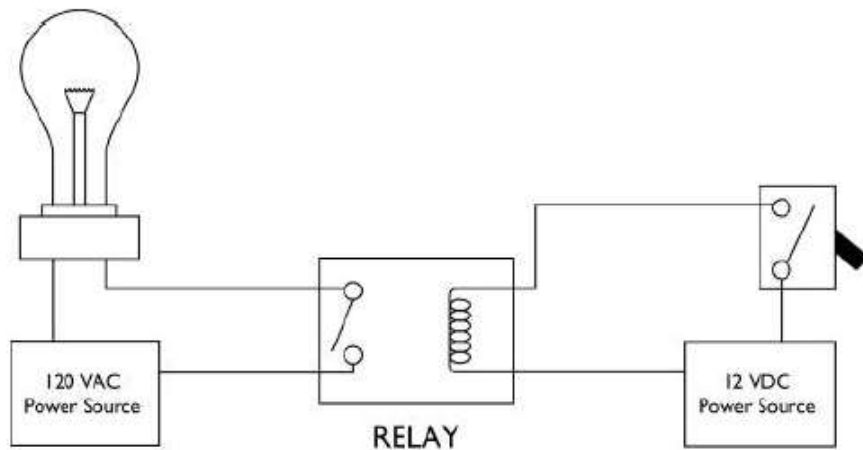
- In figure 2a the relay is off. The metal arm is at its rest position and so there is contact between the Normally Closed (N.C.) switch contact and the 'common' switch contact.
- If a current is passed through the coil, the resulting magnetic field attracts the metal arm and there is now contact between the Normally Open (N.O.) switch contact and the common switch contact, as shown in figure 2b.

Sample configuration in application



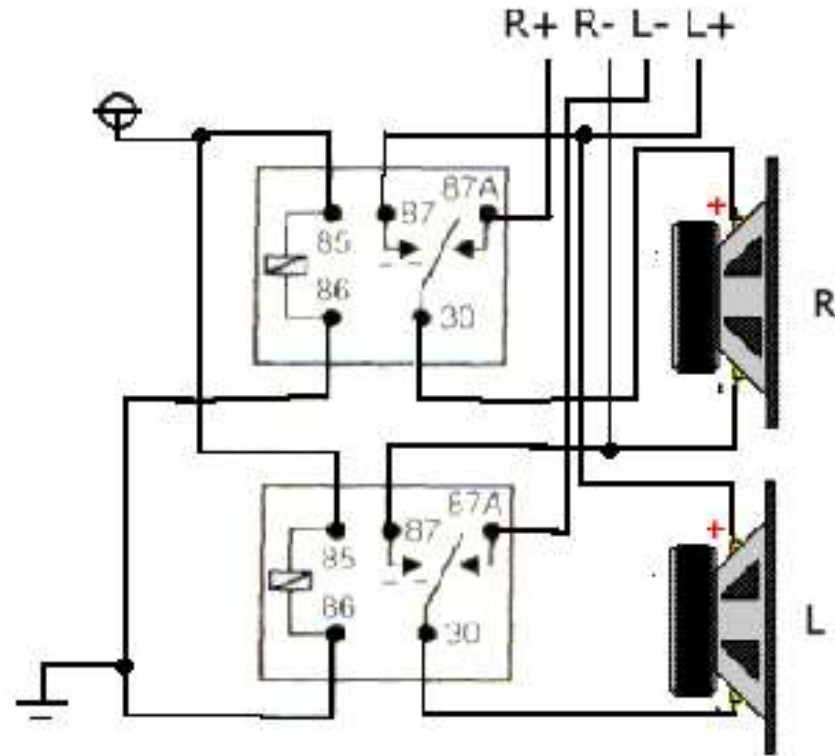
In the above schematic, the relay's coil is energized by the low-voltage (12 VDC) source, while the single-pole, single-throw (SPST) contact interrupts the high-voltage (480 VAC) circuit. The current required to energize the relay coil will be hundreds of times less than the current rating of the contact. Typical relay coil currents are well below 1 amp, while typical contact ratings for industrial relays are at least 10 amps.

Sample configuration in application



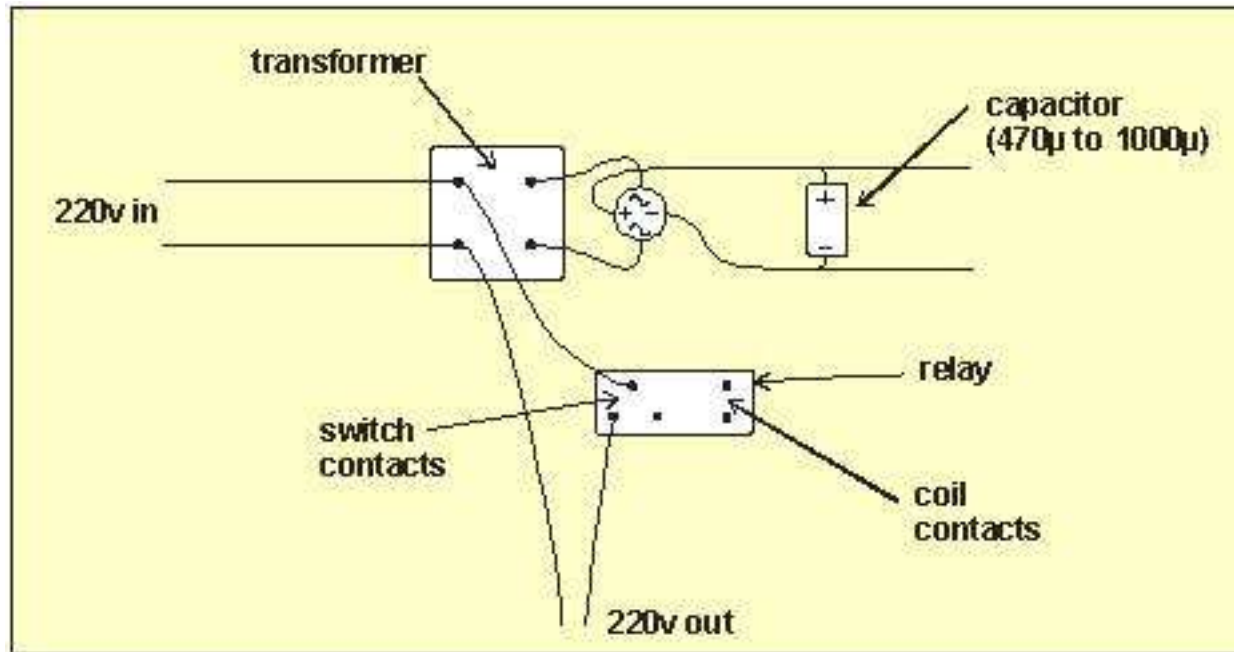
When the light switch is turned on, the relay coil is energized by the 12 VDC Power source. The coil engages the armature, which completes the path for the light bulb to get the 120 VAC from the power source.

Sample configuration in application



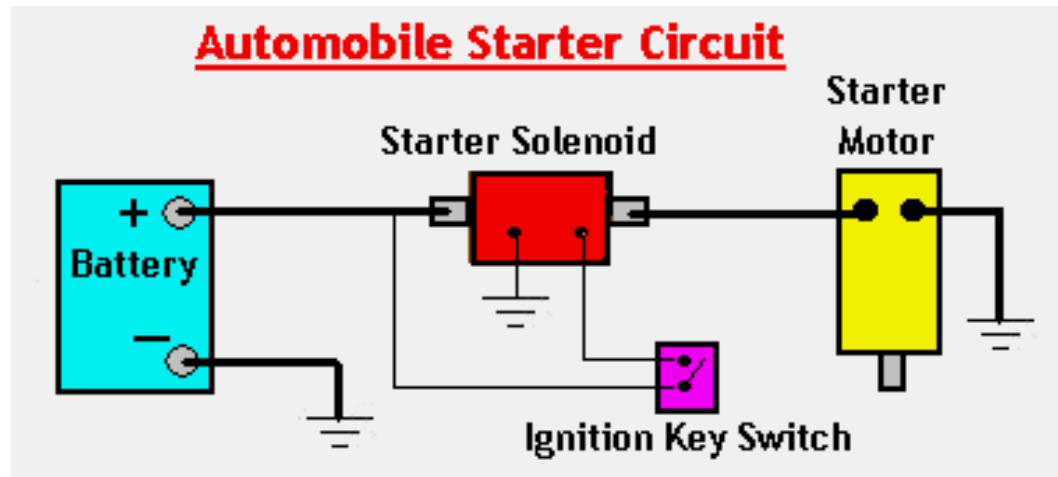
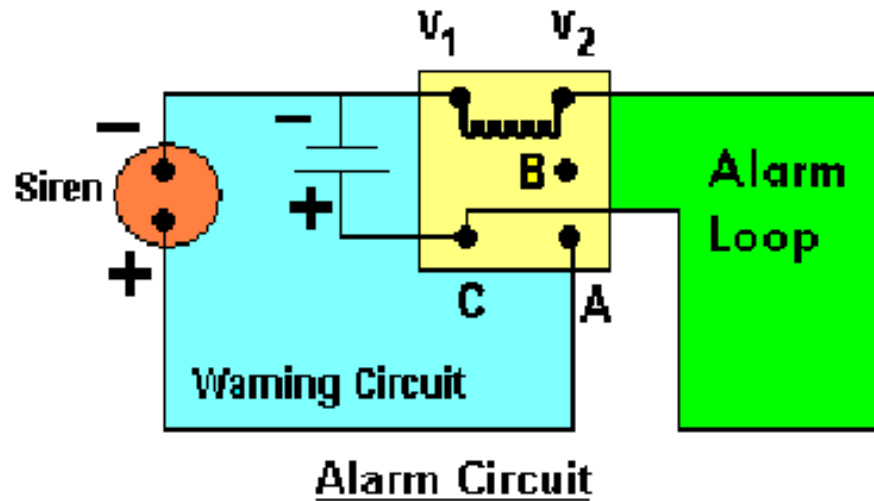
Using the Relay to select

Sample configuration in application



Using the relay as a sound operated switch

Sample configuration in application



Major specifications

- Coil Voltage
 - Pull in Voltage - minimum voltage required for the relay coil to pull the contacts together
 - Drop out Voltage – voltage at which the energized coil will release the movable contacts
- Current Carrying Capability of the contacts
 - The current rating on relay contacts tells how much current can be passed through the contacts without damage to the contacts
- Electromechanical relays have an On Resistance in the range of 100 milliohms
 - The low On Resistance of the EMR allows for greater load current capability and less signal attenuation

Major specifications

- Electromechanical relays typically have an output capacitance of less than 1 Pico Farad
- Load Type
 - Resistive Load - electric heaters, ranges and ovens, toasters and irons.
 - Inductive Loads - power drills, electric mixers, fans, sewing machines, vacuum cleaners
 - AC or DC
 - High or Low Inrush - light bulb

Limitations

- Service Life
 - At the point of the contacts, oxidation breakdown occurs over extended life cycling (typically 10^6 operations)
 - The relay will need to be replaced
- Reliability
 - When an EMR is activated, bounce occurs at the contact site
 - Bounce occurs where the load circuit is flickering between open and closed
 - Load design needs to be considered because of bounce
- Size
 - Mechanical components (coil, core, arm, contact lever arms, spring mechanism) cause physical dimension restraints
 - Package size of an EMR can limit the size of a PCB design

Limitations

- Isolation voltage
 - Most EMRs are typically rated for minimum input to output isolation voltages of 1500 to 2000 VAC.
- Speed
 - EMRs are relatively slow devices -- typical models can switch and settle in 5 to 15 ms
 - Operating speed may be too slow for some applications
- Power
 - Relays use more power due to the current flowing through their coil

Relevant issues in applications

- Physical size and pin arrangement
 - If you are choosing a relay for an existing PCB you will need to ensure that its dimensions and pin arrangement are suitable. You should find this information in the supplier's catalogue
- Coil voltage
 - The relay's coil voltage rating and resistance must suit the circuit powering the relay coil. Many relays have a coil rated for a 12V supply but 5V and 24V relays are also readily available. Some relays operate perfectly well with a supply voltage which is a little lower than their rated value
- Coil resistance
 - The circuit must be able to supply the current required by the relay coil
 - Relay coil current = supply voltage / coil resistance

Relevant issues in applications

- Switch ratings (voltage and current)
 - The relay's switch contacts must be suitable for the circuit they are to control. The voltage rating is usually higher for AC, for example: "5A at 24V DC or 125V AC"
- Relay mounting
 - Socket
 - Plug-in socket, plug-in terminal socket
 - The spade lugs of the relay can be inserted into a mating tab or into a mating socket
 - PCB mounting
 - Pins of the relay are inserted through holes in the Printed Circuit Board (PCB) designed to match the pinout of the relay and wave soldered to affix the relay to the PCB
 - Chassis
 - Mounting ears, tabs or holes are designed as part of the relays mechanical package. Those locations typically accept nuts, bolts or screws to secure the relay to some sort of chassis. This chassis may function as a mounting location only or can also be used to provide thermal management (in higher power applications).